

Digital Signal Processing and Digital Communication Systems

Assignment 4 - Advanced AI Systems

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1 Foundations of Digital Signal Processing

1.1 Introduction to Discrete-Time Signals

A discrete-time signal $x[n]$ is a sequence of numbers, often derived from a continuous-time signal $x(t)$ through sampling. According to the **Nyquist-Shannon Sampling Theorem**, to reconstruct $x(t)$ perfectly, the sampling frequency f_s must be greater than twice the maximum frequency B present in the signal: $f_s > 2B$.

1.1.1 Common Signal Sequences

1. **Unit Impulse:** $\delta[n] = 1$ for $n = 0$, else 0. 2. **Unit Step:** $u[n] = 1$ for $n \geq 0$, else 0. 3. **Exponential Sequences:** $x[n] = a^n u[n]$.

1.2 Sampling and Aliasing

When the Nyquist criterion is violated, high-frequency components are "folded" into lower frequencies, a phenomenon known as aliasing. This distortion is irreversible once the signal is sampled.

1.3 Quantization and Coding

Sampling discretizes time; quantization discretizes amplitude. A B -bit quantizer maps an infinite-precision sample to one of 2^B levels. The **Signal-to-Quantization Noise Ratio (SQNR)** for a full-scale sinusoid is approximately:

$$SQNR_{dB} \approx 6.02B + 1.76 \text{ dB} \quad (1)$$

This "6 dB per bit" rule is a fundamental design constraint in ADC (Analog-to-Digital Converter) selection.

1.4 Linear Time-Invariant (LTI) Systems

An LTI system is characterized by:

1. **Linearity:** $T\{ax_1[n] + bx_2[n]\} = aT\{x_1[n]\} + bT\{x_2[n]\}$.
2. **Time-Invariance:** If $y[n] = T\{x[n]\}$, then $y[n - k] = T\{x[n - k]\}$.

1.4.1 The Convolution Sum

The output of an LTI system is the convolution of the input $x[n]$ and the impulse response $h[n]$:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k] \quad (2)$$

1.5 Stability and Causality

Stability in the BIBO (Bounded-Input Bounded-Output) sense is crucial. An LTI system is BIBO stable if $\sum |h[n]| < \infty$. Causality implies $h[n] = 0$ for $n < 0$.

1.6 Difference Equations

Discrete-time systems are often described by linear constant-coefficient difference equations:

$$\sum_{k=0}^N a_k y[n-k] = \sum_{m=0}^M b_m x[n-m] \quad (3)$$

The order of the system is $\max(N, M)$. This representation is the basis for practical implementations of IIR and FIR filters.

1.7 The Z-Transform

The Z-Transform $X(z)$ provides a frequency-domain-like representation for discrete signals:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad (4)$$

The **Region of Convergence (ROC)** is the set of values in the complex plane for which the sum converges.

1.7.1 Properties of the Z-Transform

- **Linearity:** $ax_1[n] + bx_2[n] \leftrightarrow aX_1(z) + bX_2(z)$.
- **Time Shifting:** $x[n-k] \leftrightarrow z^{-k}X(z)$.
- **Convolution:** $x[n] * h[n] \leftrightarrow X(z)H(z)$.

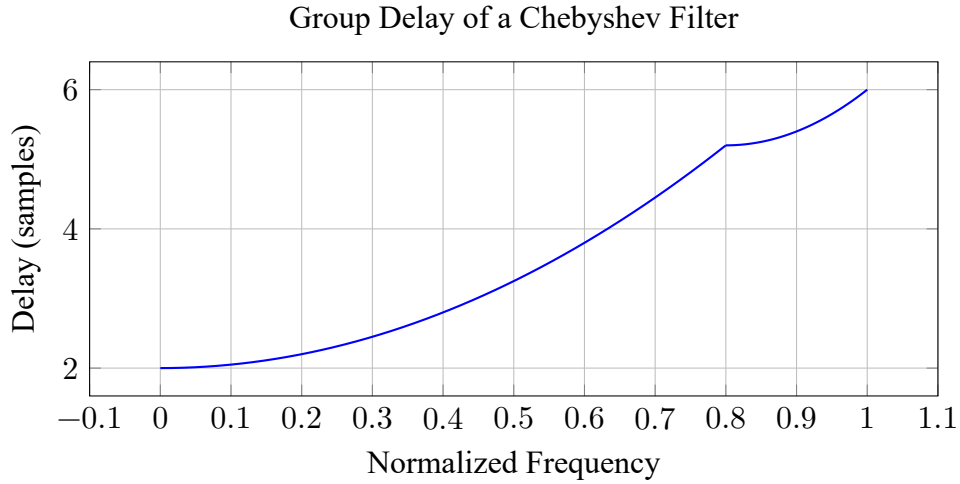
1.8 Filter Structures

Digital filters can be represented by difference equations. Porat (1997) identifies several key structures:

- **Direct Form I:** Straightforward implementation of the difference equation.
- **Direct Form II (Canonic):** Uses the minimum number of delay elements by sharing them between the feedback and feedforward paths.
- **Cascade and Parallel Forms:** Used to reduce sensitivity to coefficient quantization by breaking high-order systems into 2nd-order sections (SOS).

1.9 Group Delay and Phase Analysis

Group delay $\tau_g(\omega)$ is the negative derivative of the phase. It represents the time delay of the signal's envelope at a specific frequency. Nonlinear phase (varying group delay) causes dispersion, where different frequency components travel at different speeds.



1.10 The Hilbert Transform

The Hilbert transform is used to create the **analytic signal**, which has no negative frequency components. This is essential for SSB modulation and complex envelope representation.

$$\hat{x}[n] = x[n] * \frac{1}{\pi n} \quad (5)$$

In the frequency domain, it corresponds to a -90 degree phase shift for positive frequencies.

1.11 Multi-rate Signal Processing

Multi-rate systems use different sampling rates at various stages of processing. This is fundamental in modern transceivers for decimation (reducing f_s) and interpolation (increasing f_s).

1.11.1 Decimation and Aliasing

To reduce the sampling rate by an integer factor M , we must first apply a low-pass **anti-aliasing filter** with cutoff π/M . Then, we keep only every M -th sample:

$$y[n] = x[nM] \quad (6)$$

1.11.2 Interpolation and Imaging

To increase the sampling rate by a factor L , we insert $L - 1$ zeros between samples and then apply an **anti-imaging low-pass filter** with cutoff π/L .

1.12 Sampling of Bandpass Signals

The Nyquist theorem for bandpass signals with bandwidth B centered at f_c allows sampling at rates much lower than $2f_c$, specifically $2B < f_s < 2B(1 + 1/n)$ for some integer n , provided aliasing does not overlap the signal spectra.

2 Advanced Spectral Analysis

2.1 The Discrete Fourier Transform (DFT)

The DFT $X[k]$ is a sampled version of the Discrete-Time Fourier Transform (DTFT). It represents the signal in the discrete-frequency domain.

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} nk} \quad (7)$$

2.2 Properties of the DFT

1. **Linearity:** $DFT\{ax + by\} = aX + bY$. 2. **Periodicity:** $X[k + N] = X[k]$. 3. **Circular Shift:** $x[(n-m) \pmod{N}] \leftrightarrow X[k] e^{-j \frac{2\pi}{N} mk}$. 4. **Symmetry:** For real signals, $X[k] = X^*[N-k]$. 5. **Parseval's Relation:** $\sum |x[n]|^2 = \frac{1}{N} \sum |X[k]|^2$.

2.3 The Fast Fourier Transform (FFT)

The FFT reduces complexity from $O(N^2)$ to $O(N \log N)$.

2.3.1 Radix-2 DIT (Decimation-in-Time)

The most common algorithm, it recursively breaks an N -point DFT into two $N/2$ -point DFTs using the "butterfly" structure.

$$X[k] = G[k] + W_N^k H[k] \quad (8)$$

2.3.2 Radix-4 and Split-Radix Algorithms

For N being a power of 4, Radix-4 algorithms further reduce the number of multiplications. Split-radix combines the best of both worlds, offering even higher efficiency for specific N .

2.4 Spectral Estimation: Non-Parametric Methods

2.4.1 The Periodogram

The periodogram is a basic estimate of the Power Spectral Density (PSD). However, it is an **inconsistent estimator** because its variance does not decrease as $N \rightarrow \infty$.

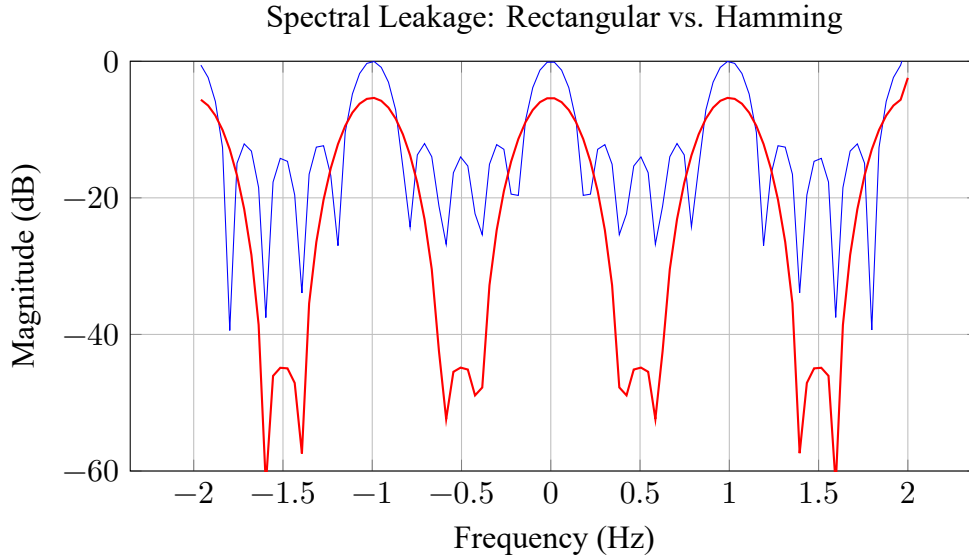
2.4.2 Welch's Method

Welch's method improves the periodogram by: 1. Dividing the signal into overlapping segments. 2. Windowing each segment. 3. Averaging the periodograms of the segments. This reduces variance at the expense of frequency resolution.

2.5 Windowing Functions

Windows are used to mitigate **spectral leakage** caused by the finite observation interval (rectangular windowing).

Window	Sidelobe Level (dB)	Main-lobe Width (bins)
Rectangular	-13	2
Hamming	-43	4
Blackman	-58	6
Kaiser ($\beta = 6$)	-50	5



2.6 Parametric Spectral Estimation

Parametric methods assume a model for the signal generation process.

2.6.1 Autoregressive (AR) Models

The signal is modeled as the output of an all-pole filter driven by white noise. The parameters are estimated using the **Yule-Walker equations** or the **Burg method**. AR models are excellent for resolving closely spaced peaks (spectral lines) in short data records.

2.6.2 Moving Average (MA) and ARMA Models

MA models are all-zero, while ARMA models have both poles and zeros. They are used for signals with broader spectral features.

2.7 High-Resolution Subspace Methods

MUSIC (Multiple Signal Classification) and **ESPRIT** exploit the eigen-structure of the auto-correlation matrix. They provide resolution far beyond the traditional DFT limit by separating the signal and noise subspaces.

2.8 Time-Frequency Analysis: The STFT

For non-stationary signals, the **Short-Time Fourier Transform (STFT)** provides a local frequency representation by sliding a window over the signal.

$$STFT\{x[n]\}(m, \omega) = \sum_{n=-\infty}^{\infty} x[n]w[n-m]e^{-j\omega n} \quad (9)$$

2.8.1 The Spectrogram

The Spectrogram is the squared magnitude of the STFT. It is a tool for visualizing non-stationary signals like speech and bird songs. The **Uncertainty Principle** dictates the trade-off between time and frequency resolution.

2.9 The Discrete Cosine Transform (DCT)

The DCT uses only cosine basis functions. It has exceptional **energy compaction** properties, making it the standard for lossy compression (JPEG, MP3).

3 Digital Filter Design and Implementation

Digital filters are fundamental components in DSP systems, used to alter the frequency spectrum of signals. This chapter provides a rigorous treatment of both Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) filter design methodologies.

3.1 Finite Impulse Response (FIR) Filters

FIR filters are characterized by an impulse response $h[n]$ of finite duration. Their transfer function is a polynomial in z^{-1} :

$$H(z) = \sum_{n=0}^{N-1} h[n]z^{-n} \quad (10)$$

3.1.1 Linear Phase Property

A key advantage of FIR filters is the ability to achieve exact linear phase, which is impossible for stable IIR filters. Linear phase ensures that all frequency components are delayed by the same amount, preserving the signal's wave shape. The condition for linear phase is symmetry or anti-symmetry:

$$h[n] = \pm h[N-1-n] \quad (11)$$

3.1.2 Window Method Design

The window method starts with the ideal (infinite) impulse response $h_d[n]$ and truncates it using a window function $w[n]$:

$$h[n] = h_d[n] \cdot w[n] \quad (12)$$

Common windows include:

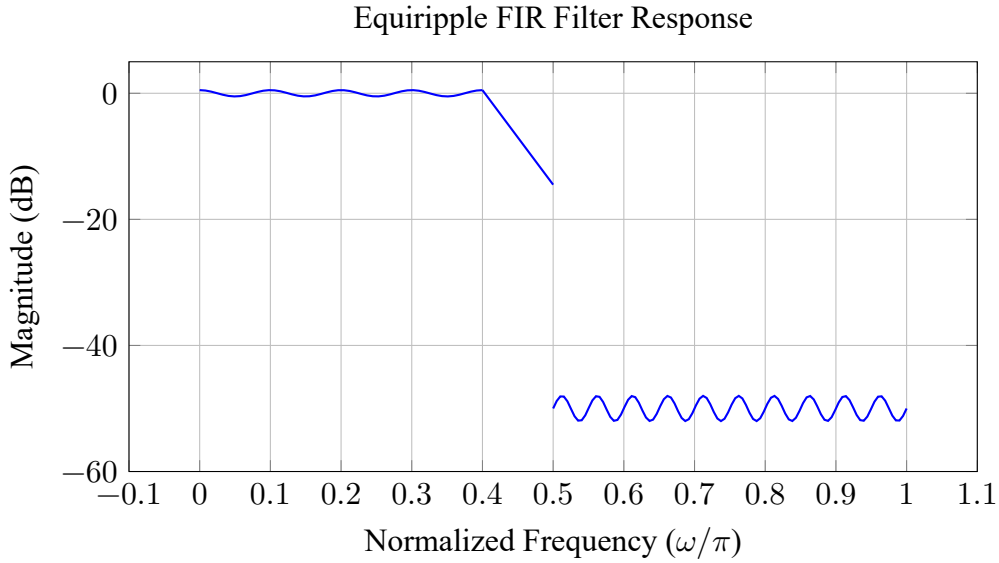
- **Rectangular:** $w[n] = 1$ for $0 \leq n \leq N-1$. High resolution but poor sidelobe rejection (-13 dB).
- **Hamming:** $w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$. Good sidelobe rejection (-43 dB).
- **Kaiser:** Allows for a trade-off between main-lobe width and sidelobe level using a parameter β .

3.1.3 Optimal FIR Design: Parks-McClellan Algorithm

The Parks-McClellan algorithm, based on the Remez exchange algorithm, produces "equiripple" filters. It minimizes the maximum error in the passband and stopband:

$$\min_{h[n]} \max_{\omega \in S} W(\omega) |H_d(e^{j\omega}) - H(e^{j\omega})| \quad (13)$$

where $W(\omega)$ is a weighting function.



3.2 Infinite Impulse Response (IIR) Filters

IIR filters use feedback, resulting in an impulse response of infinite duration. They can achieve much sharper transitions than FIR filters for the same order.

3.2.1 Classic Analog Approximations

IIR design usually begins by designing an analog prototype and then transforming it to the digital domain.

1. **Butterworth:** Maximally flat magnitude response in the passband.
2. **Chebyshev Type I:** Equiripple in the passband, monotonic in the stopband.
3. **Chebyshev Type II:** Monotonic in the passband, equiripple in the stopband.
4. **Elliptic (Cauer):** Equiripple in both passband and stopband, providing the narrowest transition band for a given order.

3.2.2 The Bilinear Transform

The Bilinear Transform is the most common method for mapping the s -plane to the z -plane. It avoids aliasing by warping the frequency axis:

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}} \quad (14)$$

The relationship between analog frequency Ω and digital frequency ω is:

$$\Omega = \frac{2}{T} \tan(\omega/2) \quad (15)$$

3.3 Filter Structures and Finite Word-Length Effects

Implementing filters in hardware or fixed-point DSPs introduces quantization errors.

- **Coefficient Quantization:** Can move poles outside the unit circle, causing instability.
- **Arithmetic Round-off Noise:** Occurs at the output of multipliers.
- **Limit Cycles:** Oscillations that can occur in IIR filters even with zero input.

IIR Filter Design via Bilinear Transform

```
import scipy.signal as signal
import numpy as np

# Design a 4th order Chebyshev Type I lowpass filter
# Passband ripple: 1dB, Cutoff frequency: 0.3*fs/2
b, a = signal.cheby1(4, 1, 0.3, btype='low', analog=False)

# Frequency response
w, h = signal.freqz(b, a)
```

3.4 Frequency Sampling Method

The frequency sampling method specifies the desired frequency response $H(e^{j\omega})$ at N equally spaced points and solves for the impulse response coefficients using the Inverse DFT:

$$h[n] = \frac{1}{N} \sum_{k=0}^{N-1} H[k] e^{j\frac{2\pi}{N}nk} \quad (16)$$

Transition band samples are often optimized to reduce ripple in the resulting continuous response.

3.5 Comparison of FIR and IIR Filters

Feature	FIR	IIR
Stability	Always Stable	Potentially Unstable
Phase	Linear Phase Possible	Non-linear Phase
Efficiency	Low (higher order)	High (lower order)
Implementation	Non-recursive	Recursive

4 Digital Communication Systems

4.1 Introduction to Digital Communication

Digital communication systems transmit discrete messages over physical channels. The main advantages over analog systems are noise immunity, ease of multiplexing, and the ability to use error correction.

4.2 Signal Space Concepts

In signal space theory, waveforms are represented as vectors in an N -dimensional space. This geometric representation simplifies the analysis of modulation performance.

4.3 Pulse Shaping and the Nyquist Criterion

To transmit symbols at a rate $1/T$ without Inter-Symbol Interference (ISI), the pulse $p(t)$ must satisfy the Nyquist Criterion.

4.3.1 Raised Cosine Filter

The Raised Cosine (RC) filter is the standard for pulse shaping. It minimizes ISI by having zero-crossings at symbol intervals.

$$P(f) = \begin{cases} T & |f| \leq \frac{1-\alpha}{2T} \\ \dots & \dots \end{cases} \quad (17)$$

where α is the **roll-off factor**.

4.3.2 Root Raised Cosine (RRC) Filter

In practical systems, the RC filter is split between the transmitter and the receiver. Each uses an RRC filter:

$$H_{TX}(f) = H_{RX}(f) = \sqrt{P(f)} \quad (18)$$

This ensures that the combined response is RC, and the receiver filter is matched to the transmitted pulse, maximizing the SNR.

4.4 Modulation Schemes

4.4.1 M-ary Phase Shift Keying (M-PSK)

Phase is modulated among M discrete values. Constant envelope makes it suitable for non-linear power amplifiers.

4.4.2 Quadrature Amplitude Modulation (QAM)

QAM varies both amplitude and phase, providing high spectral efficiency.

$$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t) \quad (19)$$

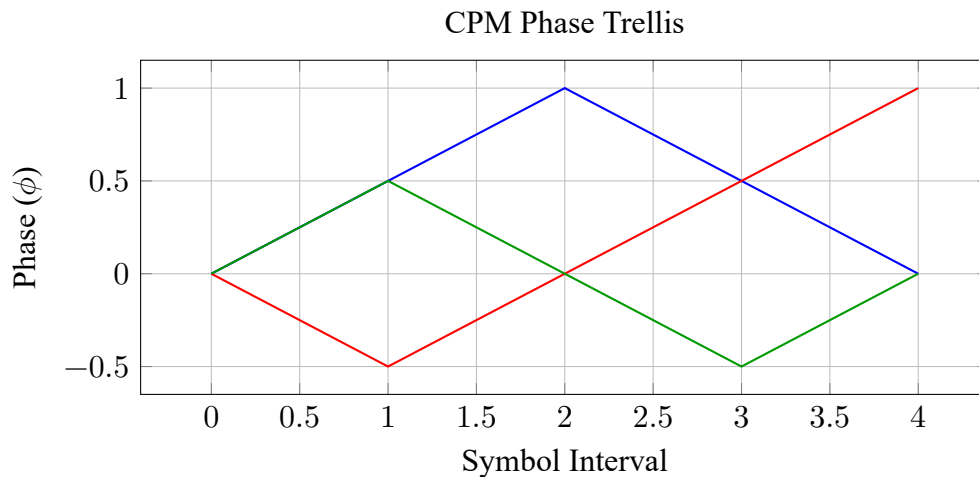
QAM is the foundation for modern high-speed wireless (Wi-Fi 6 uses 1024-QAM).

4.5 Continuous Phase Modulation (CPM)

CPM is a memory-based modulation where the phase is constrained to be continuous. This results in very low spectral sidelobes.

4.5.1 Gaussian Minimum Shift Keying (GMSK)

GMSK is a specific form of CPM where the frequency pulses are shaped by a Gaussian filter. It was the standard for GSM cellular systems due to its spectral efficiency and constant envelope.



4.6 Carrier and Symbol Synchronization

The receiver must recover the carrier frequency and phase.

4.6.1 The Costas Loop

The Costas loop is a standard method for carrier phase recovery in BPSK and QPSK. It uses two branches (I and Q) to generate an error signal that drives a Voltage Controlled Oscillator (VCO).

4.6.2 Symbol Timing Recovery

Algorithms like the **Mueller-Müller** or **Gardner** detectors are used to sample the received signal at the optimal instant.

4.7 Equalization

Equalizers combat channel-induced ISI. **Zero-Forcing (ZF)** and **Minimum Mean Square Error (MMSE)** are common linear equalizers. **Decision Feedback Equalizers (DFE)** use previous decisions to subtract ISI from the current symbol.

4.8 M-ary Modulation Comparison

The choice of modulation depends on the available bandwidth and power.

Scheme	Bits/Symbol	Bandwidth Efficiency	Power Efficiency
BPSK	1	1 bit/s/Hz	High
QPSK	2	2 bit/s/Hz	High
16-QAM	4	4 bit/s/Hz	Moderate
64-QAM	6	6 bit/s/Hz	Low

4.9 Channel Capacity: The Shannon Limit

Information theory sets the ultimate limit on how much data can be sent reliably.

$$C = B \log_2(1 + SNR) \quad (20)$$

Modern modulation and coding (like 1024-QAM with LDPC) operate within 1-2 dB of this theoretical limit.

5 Trellis Coded Modulation (TCM)

Trellis Coded Modulation, pioneered by Gottfried Ungerboeck in 1982, is a joint coding and modulation technique that achieves significant coding gains without increasing bandwidth or decreasing data rate.

5.1 The Motivation for TCM

In traditional systems, error correction coding is separate from modulation. Adding redundancy usually requires expanding the bandwidth or reducing the information rate. TCM avoids this by expanding the signal constellation and using the redundancy to constrain the allowed sequences of symbols.

5.2 Ungerboeck's Set Partitioning

The core of TCM is the principle of "mapping by set partitioning." The constellation is systematically divided into subsets with increasing intra-subset Euclidean distance.

- Start with the full constellation A_0 .
- Partition A_0 into two subsets B_0, B_1 such that the minimum distance d_{min} in B_i is greater than in A_0 .
- Continue partitioning until the desired subset distance is reached.

5.3 TCM Encoder Structure

A TCM encoder consists of a convolutional encoder followed by a mapper. For every n bits to be transmitted, the encoder might take $n - 1$ bits through a rate $1/2$ convolutional code to produce n bits, while the remaining bits are mapped directly (parallel transitions).

$$\mathbf{y} = f(\mathbf{x}, \sigma) \quad (21)$$

where σ is the state of the convolutional encoder.

5.4 Distance Properties and Coding Gain

The performance of TCM is governed by the **Free Euclidean Distance** d_{free} , which is the minimum Euclidean distance between any two distinct valid sequences of signal vectors. The asymptotic coding gain G is given by:

$$G = 10 \log_{10} \left(\frac{d_{free,TCM}^2}{d_{min,uncoded}^2} \cdot \frac{E_{avg,uncoded}}{E_{avg,TCM}} \right) \quad (22)$$

Typical TCM schemes achieve gains of 3 to 6 dB in AWGN channels.

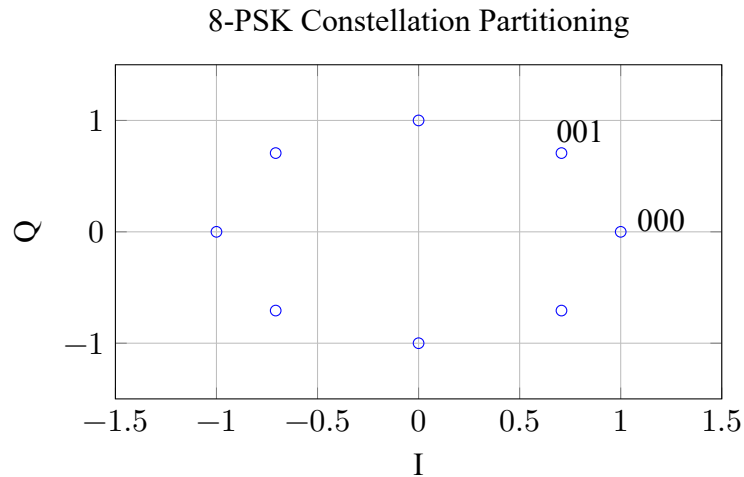
5.5 The Viterbi Decoder for TCM

The receiver uses the Viterbi algorithm to perform Maximum Likelihood Sequence Estimation (MLSE). Unlike standard convolutional codes where branch metrics are Hamming distances, TCM branch metrics are **Squared Euclidean Distances** between the received signal r_n and the constellation points s_i :

$$\lambda_n = |r_n - s_i|^2 \quad (23)$$

5.5.1 Parallel Transitions

If the trellis contains parallel transitions (representing uncoded bits), the decoder first finds the "best" point within the subset corresponding to the parallel transition (hard decision on those bits) and uses its distance as the branch metric.



5.6 Practical Applications of TCM

TCM revolutionized modem technology in the 1980s and 90s, enabling high-speed data transmission over voice-band telephone lines (ITU-T V.32, V.34 standards). It remains a cornerstone of satellite and microwave communications.

TCM Branch Metric Calculation

```
import numpy as np

def calculate_branch_metrics(received_sample, constellation):
    # Calculate Squared Euclidean Distance for all points
    distances = np.abs(received_sample - constellation)**2
    return distances

# Example for 8-PSK
received = 0.8 + 0.2j
psk8 = np.exp(1j * np.linspace(0, 2*np.pi, 8, endpoint=False))
metrics = calculate_branch_metrics(received, psk8)
```

6 Detection and Demodulation

6.1 The Optimal Receiver

In an AWGN (Additive White Gaussian Noise) channel, the optimal receiver maximizes the likelihood of the correct symbol.

6.1.1 The Matched Filter

The matched filter $h(t) = s(T - t)$ maximizes the SNR at the sampling instant. This is equivalent to a correlator receiver.

$$h(t) = s^*(T - t) \quad (24)$$

The output of the matched filter at $t = T$ is exactly the inner product $\langle r, s \rangle$.

BER Simulation for BPSK

```
import numpy as np

def simulate_ber(ebno_db, n_bits=1000000):
    ebno = 10**(ebno_db / 10)
    sigma = np.sqrt(1 / (2 * ebno))

    bits = np.random.randint(0, 2, n_bits)
    tx = 2 * bits - 1 # BPSK: 0 -> -1, 1 -> 1

    noise = sigma * np.random.randn(n_bits)
    rx = tx + noise

    # Decisions
    decisions = (rx > 0).astype(int)
    errors = np.sum(decisions != bits)
    return errors / n_bits
```

6.2 Probability of Error Analysis

The Bit Error Rate (BER) depends on the distance between constellation points and the noise power.

6.2.1 BER for BPSK

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \quad (25)$$

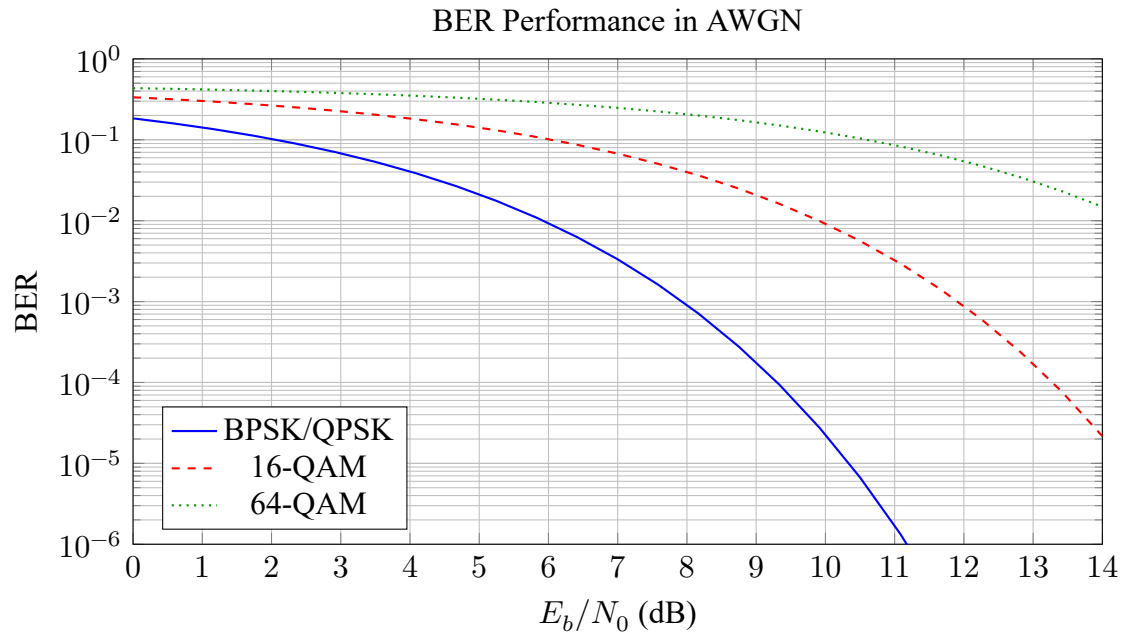
The Q-function $Q(x)$ is the tail probability of the standard normal distribution.

6.3 Coherent vs. Non-Coherent Demodulation

Coherent demodulation requires carrier phase knowledge, while non-coherent (e.g., envelope detection for FSK) does not. Coherent systems perform better but are more complex.

6.4 BER Performance Graphs

The following graph shows the theoretical BER for different modulation schemes. Note the SNR penalty for higher-order modulation.



6.5 Decision Boundaries

For an M -ary constellation, the receiver divides the I-Q plane into M decision regions. Any received point within a region is mapped to the corresponding symbol.

6.6 Maximum Likelihood Sequence Estimation (MLSE)

For channels with memory or ISI, symbol-by-symbol detection is sub-optimal. The **Viterbi Algorithm** provides the MLSE by efficiently searching the trellis of possible transmitted sequences.

6.7 Gray Coding

Gray coding ensures that adjacent constellation points differ by only one bit. This minimizes the bit error rate for a given symbol error rate.

7 Advanced Topics in Signal Processing

7.1 Multiple-Input Multiple-Output (MIMO)

MIMO systems use multiple antennas at both the transmitter and receiver to improve spectral efficiency or reliability.

7.1.1 Spatial Multiplexing

Spatial multiplexing increases the data rate by transmitting independent data streams over different spatial channels. The capacity of a MIMO system with N_t transmit and N_r receive antennas grows linearly with $\min(N_t, N_r)$.

7.1.2 Beamforming

Beamforming uses constructive and destructive interference to focus the transmitted energy in a specific direction, increasing the SNR at the receiver and reducing interference for other users.

7.2 Orthogonal Frequency Division Multiplexing (OFDM)

OFDM is the backbone of modern high-speed communication (4G, 5G, Wi-Fi). It divides a high-rate data stream into many low-rate streams transmitted on orthogonal subcarriers.

7.2.1 Cyclic Prefix (CP)

The CP is a copy of the end of an OFDM symbol prepended to the beginning. It converts linear convolution with the channel into circular convolution, simplifying frequency-domain equalization and eliminates ISI.

7.3 Spread Spectrum Communications

Spread spectrum techniques distribute the signal energy over a wide bandwidth.

7.3.1 Direct Sequence Spread Spectrum (DSSS)

In DSSS, each bit is multiplied by a high-rate **pseudo-noise (PN) sequence**. This spreads the signal's bandwidth and provides resistance to narrow-band interference and jamming.

7.3.2 Frequency Hopping Spread Spectrum (FHSS)

FHSS changes the carrier frequency rapidly according to a pre-defined sequence.

7.4 Multi-User Detection (MUD)

In multi-user systems like CDMA, the receiver can improve performance by jointly detecting all users' signals, effectively cancelling the interference they cause to each other.

Simple OFDM Symbol Generation

```
import numpy as np

def generate_ofdm_symbol(data_symbols, n_fft, cp_len):
    # IFFT
    time_domain = np.fft.ifft(data_symbols, n_fft)
    # Add Cyclic Prefix
    ofdm_symbol = np.concatenate([time_domain[-cp_len:], time_domain])
    return ofdm_symbol

# Example
n_fft = 64
cp_len = 16
data = np.random.choice([1+1j, 1-1j, -1+1j, -1-1j], n_fft)
symbol = generate_ofdm_symbol(data, n_fft, cp_len)
```

7.5 The Ambiguity Function

The ambiguity function $|\chi(\tau, f_d)|$ represents the output of the matched filter in a radar system when the received signal has a delay τ and a Doppler shift f_d . It is a critical tool for waveform design in radar engineering.

8 Case Studies and Practical Implementations

8.1 Case Study: The GSM Standard

GSM (Global System for Mobile Communications) was the first widely adopted digital cellular standard. It uses a combination of TDMA (Time Division Multiple Access) and FDMA (Frequency Division Multiple Access).

8.1.1 Modulation in GSM: GMSK

As discussed in Chapter 3, GSM uses Gaussian Minimum Shift Keying. GMSK provides a constant envelope, which is critical for efficient power amplifier operation in mobile handsets. The Gaussian filter bandwidth-time product (BT) is typically set to 0.3.

8.2 Case Study: Digital Subscriber Line (DSL)

DSL technology provides high-speed internet over traditional copper telephone lines. It uses **Discrete Multi-Tone (DMT)** modulation, which is a form of OFDM.

8.2.1 Bit Loading

DMT adaptively allocates bits to different subcarriers based on the Signal-to-Noise Ratio (SNR) of each subchannel. This allows DSL to maximize the data rate even in the presence of frequency-selective impairments like bridge taps.

8.3 Case Study: 5G New Radio (NR)

5G NR introduces several advanced features to support massive machine-type communications (mMTC) and ultra-reliable low-latency communications (URLLC).

8.3.1 Massive MIMO and Beamforming

5G uses large antenna arrays (up to 256 elements) to perform highly directional beamforming. This increases the signal strength at the intended user while reducing interference for others.

8.4 Case Study: Evolution of IEEE 802.11 (Wi-Fi)

The Wi-Fi standard has evolved from 2 Mbps (802.11b) to multi-Gbps rates (802.11ax/be) by increasing bandwidth, using higher-order modulation (1024-QAM), and employing massive MIMO.

8.4.1 Wi-Fi 6 (802.11ax)

Wi-Fi 6 introduced **OFDMA** (Orthogonal Frequency Division Multiple Access), allowing the router to serve multiple clients simultaneously within a single symbol interval by allocating subcarrier subsets (Resource Units) to different users.

8.5 Practical Implementation: FPGA-based Digital Down-Converter (DDC)

A DDC is used in modern receivers to shift a high-frequency IF signal to baseband and reduce the sampling rate.

1. **Numerical Control Oscillator (NCO):** Generates a complex sinusoid.
2. **Mixer:** Performs the frequency shift.
3. **CIC Filter:** Efficiently performs decimation and initial filtering.
4. **FIR Decimation Filter:** Provides sharp filtering and further decimation.

Simple Digital Down-Conversion

```
import numpy as np
from scipy import signal

# IF Signal at 10 MHz, Sampled at 100 MHz
fs = 100e6
t = np.arange(1000) / fs
x_if = np.cos(2*np.pi*10e6*t)

# NCO at 10 MHz
nco = np.exp(-1j * 2*np.pi*10e6*t)

# Mixing to Baseband
x_baseband = x_if * nco

# Low-pass filter and Decimate by 10
b = signal.firwin(64, 1e6, fs=fs)
x_lowpass = signal.lfilter(b, 1, x_baseband)
x_final = x_lowpass[::10]
```

9 Radar Signal Processing

Digital Signal Processing is fundamental to modern radar systems. This chapter explores the principles of pulse compression, Doppler processing, and the ambiguity function.

9.1 Basic Radar Principles

A radar system transmits a signal $s(t)$ and listens for echoes. The time delay Δt between transmission and reception determines the range R :

$$R = \frac{c\Delta t}{2} \quad (26)$$

where c is the speed of light.

9.2 Pulse Compression

To achieve high range resolution without sacrificing transmit energy, radar systems use long, modulated pulses rather than short, high-power pulses.

9.2.1 Linear Frequency Modulation (LFM) - Chirp

The LFM signal, or chirp, increases its frequency linearly over the pulse duration T :

$$s(t) = e^{j\pi K t^2}, \quad -T/2 \leq t \leq T/2 \quad (27)$$

where K is the chirp rate. At the receiver, a matched filter (pulse compressor) collapses the pulse to a width of approximately $1/B$, where $B = KT$ is the bandwidth.

9.3 The Ambiguity Function

The ambiguity function $|\chi(\tau, f_d)|$ represents the output of the matched filter when the received signal has a delay τ and a Doppler shift f_d . It defines the resolution and clutter rejection capabilities of a specific waveform.

9.4 Doppler Processing and MTI

Moving Target Indication (MTI) uses the Doppler shift $f_d = \frac{2v}{\lambda}$ to distinguish moving targets from stationary ground clutter. Modern systems use **Pulse Doppler Radar** with FFT-based processing to resolve targets in both range and velocity.

LFM Chirp Generation

```
import numpy as np
import matplotlib.pyplot as plt

T = 10e-6      # 10 microseconds
B = 5e6        # 5 MHz
fs = 50e6      # 50 MHz
t = np.arange(-T/2, T/2, 1/fs)
K = B/T
chirp = np.exp(1j * np.pi * K * t**2)
```

10 Information Theory and Source Coding

Information theory, founded by Claude Shannon in 1948, provides the mathematical framework for understanding the limits of data compression and reliable communication.

10.1 Measure of Information: Entropy

Entropy $H(X)$ measures the average uncertainty associated with a random variable X . For a discrete random variable with probability mass function $p(x)$, entropy is defined as:

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log_2 p(x) \quad [\text{bits}] \quad (28)$$

Entropy is maximized when all outcomes are equally likely.

10.2 Mutual Information and Channel Capacity

Mutual information $I(X; Y)$ measures the amount of information that one random variable contains about another. It is the reduction in uncertainty of X due to the knowledge of Y :

$$I(X; Y) = H(X) - H(X|Y) \quad (29)$$

The **Channel Capacity** C is the maximum rate at which information can be transmitted reliably over a channel:

$$C = \max_{p(x)} I(X; Y) \quad (30)$$

10.2.1 The Shannon-Hartley Theorem

For an AWGN channel with bandwidth B and signal-to-noise ratio SNR , the capacity is:

$$C = B \log_2(1 + SNR) \quad [\text{bits/s}] \quad (31)$$

This theorem sets an absolute limit on communication performance. No matter how clever the coding, reliable communication is impossible if the rate $R > C$.

10.3 Source Coding (Data Compression)

Source coding aims to represent information as efficiently as possible by removing redundancy.

10.3.1 Huffman Coding

Huffman coding is an optimal prefix-free code for a given set of symbol probabilities. It assigns shorter codes to more frequent symbols. *Example Algorithm:*

1. List symbols and their probabilities.
2. Combine the two least probable symbols into a new node.
3. Repeat until a single root node remains.

10.3.2 Lempel-Ziv (LZ77/LZ78) Coding

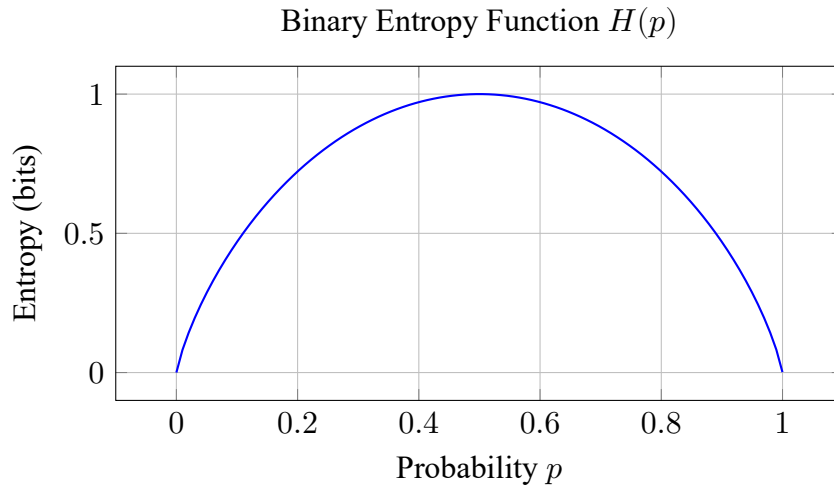
Lempel-Ziv algorithms are universal source codes that do not require prior knowledge of symbol probabilities. They work by building a dictionary of previously seen patterns. LZ coding is the basis for many modern compression formats like ZIP, GZIP, and PNG.

10.4 Differential Entropy and Gaussian Channels

For continuous random variables, the entropy is replaced by differential entropy $h(X)$:

$$h(X) = - \int f(x) \log_2 f(x) dx \quad (32)$$

For a given variance, the Gaussian distribution has the maximum differential entropy. This property is fundamental to the derivation of the AWGN channel capacity.



10.5 Rate-Distortion Theory

Rate-distortion theory deals with lossy compression. it defines the minimum rate $R(D)$ required to represent a source with a maximum allowable distortion D . This is the theoretical foundation for image (JPEG) and video (MPEG) compression.

11 Channel Coding and Error Control

Channel coding introduces structured redundancy to detect and correct errors caused by noise and interference in the communication channel.

11.1 Linear Block Codes

A linear block code (n, k) maps k information bits into n coded bits. It is defined by its generator matrix $\mathbf{G}$:

$$\mathbf{c} = \mathbf{u}\mathbf{G} \quad (33)$$

The parity-check matrix $\mathbf{H}$ is used to verify the validity of received codewords: $\mathbf{c}\mathbf{H}^T = \mathbf{0}$.

11.1.1 Hamming Distance and Error Correction

The minimum distance d_{min} of a code determines its error-control capability:

- Can detect up to $d_{min} - 1$ errors.
- Can correct up to $\lfloor (d_{min} - 1)/2 \rfloor$ errors.

Hamming Codes are a class of $(2^m - 1, 2^m - 1 - m)$ codes with $d_{min} = 3$, capable of correcting a single error.

11.2 Cyclic Codes and CRC

Cyclic codes are a subclass of linear block codes where any circular shift of a codeword is also a codeword. They are efficiently implemented using shift registers and are the basis for **Cyclic Redundancy Checks (CRC)**, used for error detection in almost all networking protocols.

11.3 Convolutional Codes

Unlike block codes, convolutional codes have memory. The output at any time depends on the current and previous input bits. They are characterized by their constraint length K and code rate R .

11.3.1 The Viterbi Algorithm

The Viterbi algorithm is the optimal (ML) decoding method for convolutional codes. It finds the most likely path through the code trellis by recursively calculating and pruning paths based on accumulated metrics (Hamming distance for hard decisions, Euclidean for soft).

11.4 Modern High-Performance Codes

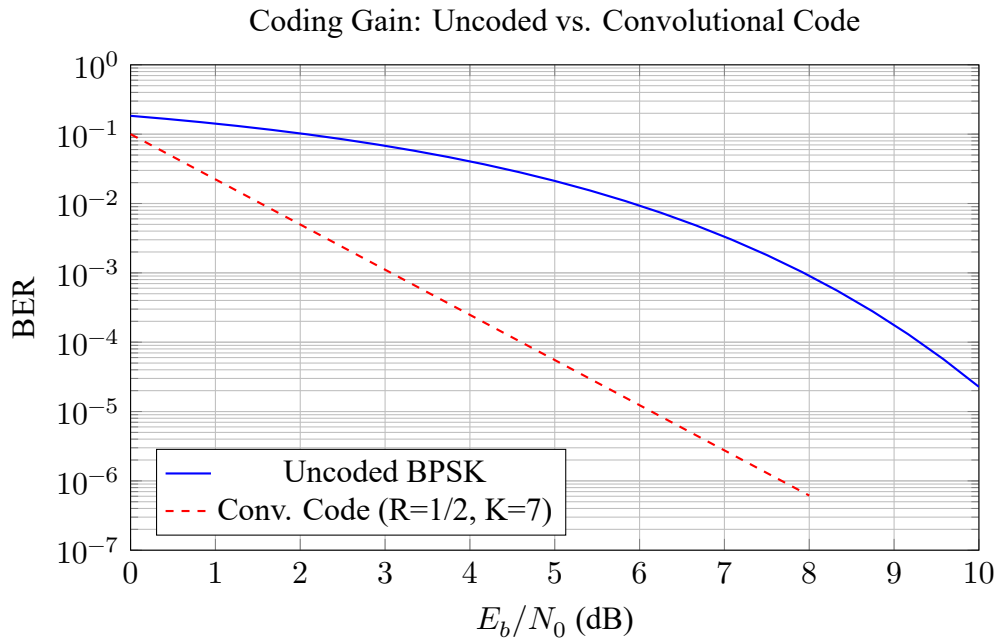
Traditional codes have a gap between their performance and the Shannon limit. Modern codes bridge this gap.

11.4.1 Turbo Codes

Introduced in 1993, Turbo codes use parallel concatenated convolutional codes and iterative "turbo" decoding. They were the first practical codes to approach the Shannon limit and were adopted in 3G and 4G standards.

11.4.2 Low-Density Parity-Check (LDPC) Codes

Originally proposed by Gallager in 1962 but rediscovered in the 1990s, LDPC codes use sparse parity-check matrices and the **Belief Propagation** algorithm for decoding. They are highly parallelizable and are used in 5G NR and high-speed Wi-Fi (802.11n/ac/ax).



11.5 Soft-Decision Decoding

Soft-decision decoding uses the analog values from the demodulator (rather than 0/1 bits) to calculate metrics. This provides a gain of approximately 2-3 dB over hard-decision decoding in AWGN channels.

12 Wireless Communication and Fading Channels

The wireless channel is far more challenging than the stationary AWGN channel due to mobility and multipath propagation.

12.1 Multipath Propagation and Fading

In a wireless environment, the signal reaches the receiver via multiple paths (reflection, diffraction, scattering). These paths have different delays, phases, and amplitudes.

12.1.1 Rayleigh Fading

When there is no line-of-sight (LOS) component, the received signal envelope follows a Rayleigh distribution. This is typical in dense urban environments.

12.1.2 Ricean Fading

When a dominant LOS path exists, the envelope follows a Ricean distribution, characterized by the K -factor (ratio of LOS power to multipath power).

12.2 Small-Scale Fading Parameters

- **Delay Spread:** The difference in arrival times of the paths. It leads to **frequency-selective fading** if the bandwidth exceeds the coherence bandwidth.
- **Doppler Spread:** Caused by mobility, leading to frequency shifts. It causes **time-selective fading** if the symbol duration exceeds the coherence time.

12.3 Diversity Techniques

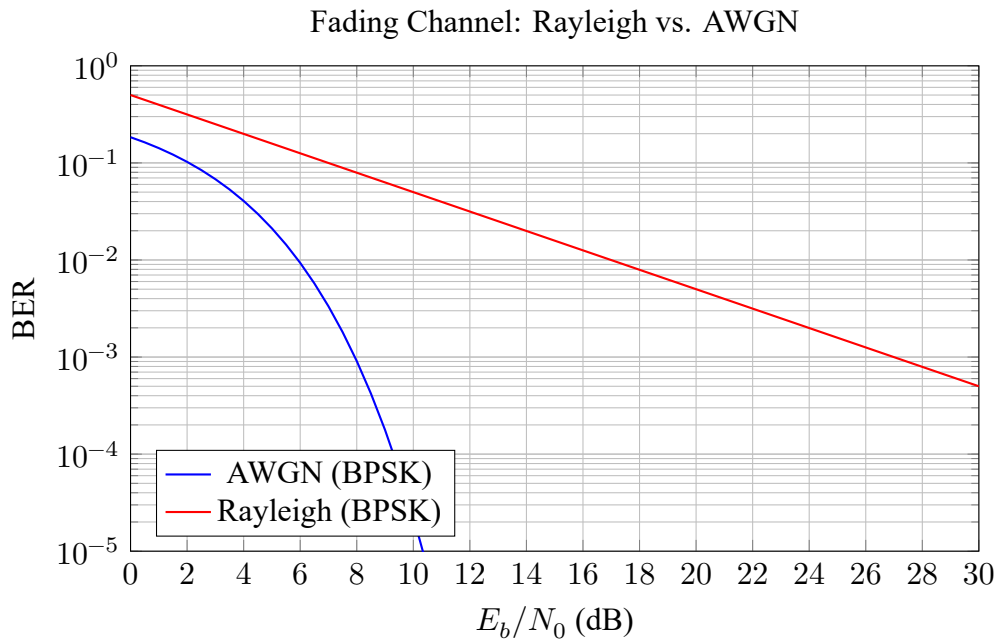
Diversity is the primary tool to combat fading by providing multiple independent copies of the signal.

1. **Temporal Diversity:** Interleaving and coding.
2. **Frequency Diversity:** OFDM and spread spectrum.
3. **Spatial (Antenna) Diversity:** Using multiple antennas (MIMO).

12.3.1 MIMO: Multiple-Input Multiple-Output

MIMO systems use multiple antennas at the transmitter and receiver.

- **Spatial Multiplexing:** Increases data rate by sending independent streams.
- **Beamforming:** Focuses energy in a specific direction to increase SNR.
- **Space-Time Coding:** Increases reliability (e.g., Alamouti scheme).



12.4 Channel Estimation and Equalization

In fading channels, the receiver must estimate the channel's complex gain (CSI - Channel State Information) to perform coherent demodulation. Pilots (known symbols) are periodically transmitted for this purpose. Adaptive equalizers then invert the channel's impact.

13 Adaptive Filtering and Signal Processing

Adaptive filters are filters whose coefficients adjust themselves to minimize a cost function, typically the Mean Squared Error (MSE). They are essential for non-stationary environments.

13.1 The Wiener Filter

The Wiener filter is the optimal linear filter for estimating a target signal $d[n]$ from an observed signal $x[n]$ in the sense of minimizing the MSE. The optimal coefficients $\mathbf{w}_{opt}$ are given by the **Wiener-Hopf equations**:

$$\mathbf{R}_x \mathbf{w}_{opt} = \mathbf{p} \quad (34)$$

where $\mathbf{R}_x$ is the autocorrelation matrix of the input and $\mathbf{p}$ is the cross-correlation vector.

13.2 The Least Mean Square (LMS) Algorithm

The LMS algorithm is a stochastic gradient descent method that iteratively updates the filter weights without requiring explicit calculation of $\mathbf{R}_x$. *Weight Update Equation*:

$$\mathbf{w}[n+1] = \mathbf{w}[n] + \mu e[n] \mathbf{x}[n] \quad (35)$$

where $e[n] = d[n] - \mathbf{w}^H[n] \mathbf{x}[n]$ is the error signal and μ is the step size. LMS is widely used due to its computational simplicity $O(N)$.

13.3 The Recursive Least Squares (RLS) Algorithm

The RLS algorithm minimizes a weighted sum of squared errors. It offers much faster convergence than LMS, especially in environments with high eigenvalue spread in $\mathbf{R}_x$, but at the cost of higher complexity $O(N^2)$.

$$J[n] = \sum_{i=0}^n \lambda^{n-i} |e[i]|^2 \quad (36)$$

where λ is the forgetting factor.

13.4 Applications of Adaptive Filters

13.4.1 Adaptive Equalization

Used in communication receivers to mitigate ISI caused by time-varying channels. The equalizer "learns" the inverse of the channel.

13.4.2 Acoustic Echo Cancellation (AEC)

Used in speakerphones and video conferencing to remove the echo of the far-end speaker's voice from the local microphone signal.

13.4.3 Active Noise Control

Used in noise-canceling headphones to generate "anti-noise" that cancels out ambient environmental sounds.

LMS Algorithm Implementation

```
import numpy as np

def lms_filter(x, d, N, mu):
    w = np.zeros(N)
    y = np.zeros(len(x))
    e = np.zeros(len(x))
    for n in range(N, len(x)):
        x_n = x[n:n-N:-1]
        y[n] = np.dot(w, x_n)
        e[n] = d[n] - y[n]
        w = w + mu * e[n] * x_n
    return y, e, w
```

13.5 Kalman Filtering

The Kalman filter is an optimal recursive estimator for the state of a dynamic system. It is a generalization of the Wiener filter to state-space models and is fundamental to radar tracking, navigation (GPS/INS), and control systems.

14 Modern Communication Standards and Future Trends

This chapter reviews the implementation of high-performance DSP and communication theories in contemporary and upcoming standards.

14.1 5G New Radio (NR)

5G NR is designed for high throughput (eMBB), low latency (URLLC), and massive connectivity (mMTC).

- **Scalable OFDM:** Flexible subcarrier spacing to handle different frequency bands (from sub-6GHz to mmWave).
- **Massive MIMO:** Hundreds of antennas performing 3D beamforming to serve multiple users in the same frequency-time resource (MU-MIMO).
- **Advanced Coding:** LDPC for data channels and Polar codes for control channels.

14.2 Wi-Fi 7 (IEEE 802.11be)

Wi-Fi 7 is the upcoming standard for "Extremely High Throughput" (EHT).

- **320 MHz Bandwidth:** Doubling the channel size from Wi-Fi 6.
- **4096-QAM (4K-QAM):** Increasing the number of bits per symbol to 12.
- **Multi-Link Operation (MLO):** Allowing a device to transmit and receive across multiple bands (2.4, 5, and 6 GHz) simultaneously.

14.3 Internet of Things (IoT) Connectivity

14.3.1 LoRa (Long Range)

LoRa uses **Chirp Spread Spectrum (CSS)** modulation to achieve extremely long ranges (up to 15km) with very low power consumption, albeit at low data rates. It is ideal for battery-powered sensors.

14.3.2 NB-IoT (Narrowband IoT)

NB-IoT is a cellular-based standard that uses a single 180 kHz PRB (Physical Resource Block) of the LTE/5G spectrum. It provides deep indoor coverage and supports massive numbers of low-throughput devices.

14.4 Satellite Communications: Starlink and Beyond

Modern LEO (Low Earth Orbit) satellite constellations use advanced phased arrays and beamforming to provide high-speed internet globally. They employ high-order modulation and sophisticated interference mitigation to reuse spectrum across large geographic areas.

14.5 The Path to 6G

While 5G is still being deployed, research into 6G has begun. Expected features include:

- **Terahertz (THz) Communication:** Massive bandwidth at 100 GHz - 3 THz.
- **Joint Communications and Sensing (JCAS):** Using the network as a high-resolution radar.
- **AI-Native Air Interface:** Using deep learning to optimize the entire physical layer, replacing traditional DSP algorithms.

Feature	4G (LTE)	5G NR	6G (Goal)
Peak Rate	1 Gbps	20 Gbps	1 Tbps
Latency	10-50 ms	1 ms	0.1 ms
Mobility	350 km/h	500 km/h	1000 km/h

A Mathematical Appendices

A.1 Fourier Transform Properties

Property	Time Domain $x(t)$	Frequency Domain $X(f)$
Duality	$X(t)$	$x(-f)$
Scaling	$x(at)$	$\frac{1}{ a } X(f/a)$
Differentiation	$\frac{d^n}{dt^n} x(t)$	$(j2\pi f)^n X(f)$
Integration	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{j2\pi f} X(f) + \dots$

A.2 Probability Distributions in Communication

A.2.1 Gaussian (Normal) Distribution

Describes the noise in AWGN channels:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (37)$$

A.2.2 Rayleigh Distribution

Describes the envelope of a received signal in a non-line-of-sight (NLOS) fading environment.

A.2.3 Rician Distribution

Describes the envelope when there is a strong line-of-sight (LOS) component in addition to multipath fading.

A.3 Common Identities and Sums

1. **Geometric Series:** $\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a}$. 2. **Euler's Formula:** $e^{j\theta} = \cos \theta + j \sin \theta$. 3. **Parseval's Theorem:** Total energy in time domain equals total energy in frequency domain.

A.4 Derivation of the Matched Filter SNR

The Matched Filter $H(f)$ maximizes the peak instantaneous SNR:

$$\eta = \frac{|y(T)|^2}{E[n^2(t)]} \quad (38)$$

By applying the Schwarz Inequality to the output signal $y(T) = \int S(f)H(f)e^{j2\pi fT} df$, we find that the maximum occurs when $H(f) = kS^*(f)e^{-j2\pi fT}$, which in the time domain is $h(t) = ks^*(T-t)$.

References